



International Journal of Applied Earth Observation
and Geoinformation xxx (2008) xxx–xxx

INTERNATIONAL JOURNAL OF
APPLIED EARTH OBSERVATION
AND GEOINFORMATION

www.elsevier.com/locate/jag

Crop discrimination in Northern China with double cropping systems using Fourier analysis of time-series MODIS data

Zhang Mingwei ^{a,*}, Zhou Qingbo ^b, Chen Zhongxin ^{b,1}, Liu Jia ^b,
Zhou Yong ^c, Cai Chongfa ^d

^a National Satellite Meteorological Center, China Meteorology Administration, No. 46 Zhongguancun South Avenue,
Beijing 100081, China

^b Institute of Natural Resources & Regional Planning, The Chinese Academy of Agricultural Sciences,
No. 12 Zhongguancun South Avenue, Beijing 100081, China

^c Huazhong Normal University, No. 152 Luoyu Road, Wuhan, Hubei 430079, China

^d Huazhong Agricultural University, Wuhan, Hubei 430070, China

Received 28 June 2006; accepted 14 November 2007

Abstract

Crop identification is the basis of crop monitoring using remote sensing. Remote sensing the extent and distribution of individual crop types has proven useful to a wide range of users, including policy-makers, farmers, and scientists. Northern China is not merely the political, economic, and cultural centre of China, but also an important base for grain production. Its main grains are wheat, maize, and cotton. By employing the Fourier analysis method, we studied crop planting patterns in the Northern China plain. Then, using time-series EOS-MODIS NDVI data, we extracted the key parameters to discriminate crop types. The results showed that the estimated area and the statistics were correlated well at the county-level. Furthermore, there was little difference between the crop area estimated by the MODIS data and the statistics at province-level. Our study shows that the method we designed is promising for use in regional spatial scale crop mapping in Northern China using the MODIS NDVI time-series.

© 2007 Elsevier B.V. All rights reserved.

Keywords: MODIS; NDVI; Crop discrimination; Fourier transform

1. Introduction

Crop identification is the basis for crop monitoring using remote sensing and is critical to many applications. For example, crop acreage is essential information necessary for land management and trade decisions.

Identification and mapping of crop types using coarse-resolution satellite imagery is an important step to assure accurate retrieval of crop specific parameters (Doraiswamy et al., 2005). Crop discrimination using low-resolution satellite imagery is a critical component of mesoscale storm prediction (Gutman and Ignatov, 1998) and hydrologic models (Yin, 1997).

Vegetation types can be characterized using their seasonal (or phenological) variations in the NDVI time-series (Townshend et al., 1991). During the last decade, a number of different methods have been developed to discriminate crop types using data from the normalized difference vegetation index (NDVI) and from the Advanced Very High-Resolution Radiometer (AVHRR).

* Corresponding author. Tel.: +86 10 6840 6707;
fax: +86 10 6217 5936.

E-mail addresses: zmw0910@gmail.com (Z. Mingwei),
zhouqb@mail.caas.net.cn (Z. Qingbo), zxcchen@mail.caas.net.cn
(C. Zhongxin).

¹ Tel.: +86 10 6891 9615x168/6891 8684;
fax: +8610 6891 9615x150.

These methods employ a variety of different approaches including temporal profiles of crop phenology manifested in the normalized difference vegetation index (Defries et al., 1995; Reed et al., 1994), classification of multi-temporal data (Brown et al., 1993; Loveland et al., 1995), the Principal Component Analysis (Townshend et al., 1987; Tucher et al., 1985), and time-series analysis of a temporal NDVI profile as a standardized principal component analysis (Eastman and Fulk, 1993).

Fourier analysis provides a new representation of the time-series of images, which allows analysis of the vegetation phenology using only the amplitude and phase of the most important periodic components. The approach by Menenti et al. (1993) provides a measure of the ‘quality’ of vegetation zones in dynamic terms using the FFT: the amplitude and phase values at different frequencies are a measure of the response of vegetation to different periodic climate processes. Images of amplitude and phase values were used as attributes to map vegetation–soil–climate units in Southern Africa (Azzali and Menenti, 2000). Amplitude and phase angle images were correlated with crop type information (Jakubauskas et al., 2001). Jakubauskas et al. (2002) used amplitude and phase angle images to discriminate crops. Finally, Verhoef (Roerink and Menenti, 2000) developed the algorithm Harmonic Analysis of Time-Series (HANTS) to deal with time-series of irregularly spaced observations and to identify and remove cloud contaminated observations. HANTS has to be steered by five control parameters. There are no objective rules to determine the magnitude of these control parameters.

The radiometric and geometric properties of the Moderate Resolution Imaging Spectroradiometer (MODIS), in combination with improved atmospheric correction and cloud screening provided by MODIS science team activities, provide a substantially improved basis for studies of this nature (Zhang et al., 2003). The MODIS sensor has 36 spectral bands, seven of which are designed for the study of vegetation and land surfaces: blue (459–479 nm), green (545–565 nm), red (620–670 nm), near infrared (NIR1: 841–875 nm, NIR2: 1230–1250 nm), and shortwave infrared (SWIR1: 1628–1652 nm, SWIR2: 2105–2155 nm). Daily global imagery is provided at spatial resolutions of 250 m (red and NIR1) and 500 m (blue, green, NIR2, SWIR1, SWIR2). The MODIS Land Science Team provides a suite of standard MODIS data products to users, including the 8-day composite MODIS Surface Reflectance Product (MOD09A1). Among a suite of standard MODIS data products available to users, we used the 8-day composite MODIS Surface Reflectance Product

(MOD09A1). Each 8-day composite includes estimates of surface spectral reflectance of the seven spectral bands at 500-m spatial resolution. Compared to other MODIS data products, the 8-day composite MODIS Surface Reflectance Product (MOD09Q1) has three advantages for crop analyses: (a) finer spatial resolution (500 m versus 1 km in Vegetation Indices 16-Day L3 Global 1 km (MOD13A2)), (b) slightly shorter temporal resolution (8-day versus 16-day in Vegetation Indices 16-Day L3 Global 250 m (MOD13Q1)), (c) more bands (7-band versus 2-band in Surface Reflectance 8-Day L3 Global 250 m (MOD09Q1)).

Several studies on crop discrimination using NDVI time-series refer mostly to areas in North America, Europe and elsewhere, few reports discuss double cropping systems in northern China. The objective of this study is to estimate crop acreage in the northern China plain with double cropping systems using 500 m, 8-day composite MODIS NDVI. In this paper, the Fast Fourier Transform (FFT) is employed to extract phase and amplitude images from MODIS-NDVI time-series data. The phase and amplitude images are used to discriminate crop types.

2. Study area

The research region is located in northern China (30–43E, 110–123W), including Beijing and Tianjin municipalities, and the Hebei, Shandong, Henan and Shandong provinces, which is China’s food production base (Fig. 1). The region’s area is approximately 539,508 km². Crop land dominates the relatively flat landscape of the North China Plain, comprising about 62.91% of the total area. Dry-land-agriculture dominates the North China Plain, with winter wheat, maize, and cotton as the principal crops.

Multiple cropping systems are mainly practiced in the southern areas of northern China, where there is a temperate monsoon climate, sufficient rainfall, and a long frost-free period. The rotation of winter wheat–maize and the relay intercropping of winter wheat–cotton is generally practiced in areas where a double cropping system is used. In northern areas within the North China Plain, the one cropping system is mainly practiced because of the cold climate and short frost-free period. The spring maize and cotton are the main crops in the area with a one-crop-a-year-system. According to the crop calendar of northern China, which was obtained from agro-meteorological stations (Fig. 2), cotton’s growth period is longer than that of maize, and it ends in October.

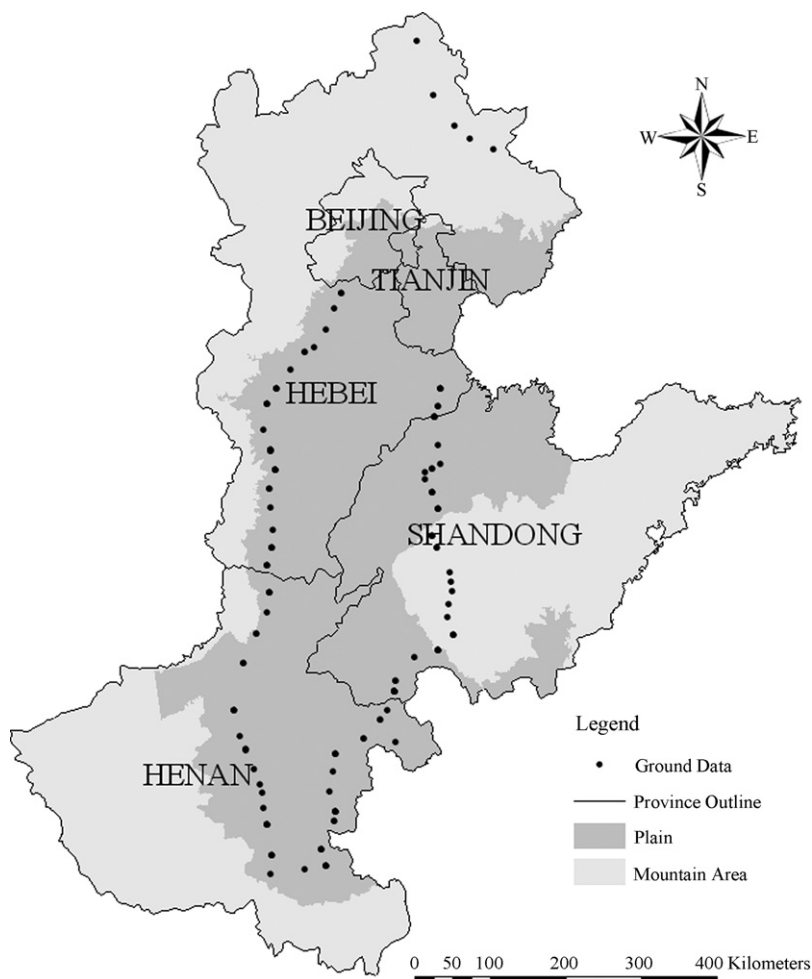


Fig. 1. Study area and survey sites.

3. Data and data sets

3.1. MODIS data sets

The MODIS data were provided by the EOS data Gateway (EOS, 2005). We used the 8-day composite MODIS Surface Reflectance Product (MOD09A1). Each

8-day composite includes estimates of ground spectral reflectance of the seven spectral bands at 500 m spatial resolution. The product was composed so as to have the lowest value of blue band (band 3) over every 8 days. In addition, the MOD09A1 product implements corrections for gaseous, aerosol scattering, and thin cirrus (Vermote and Vermeulen, 1999). Even with correction, the noise,

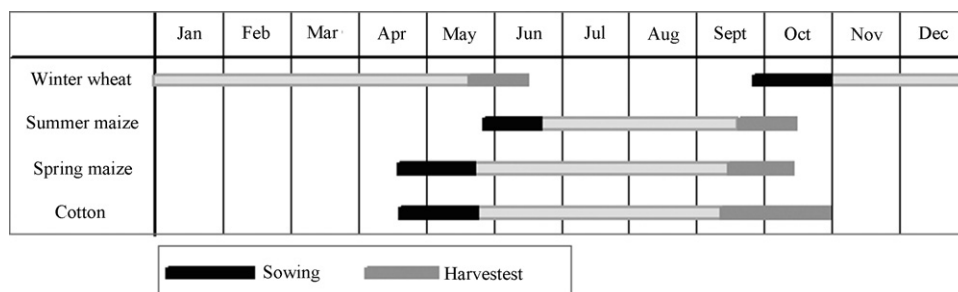


Fig. 2. Crop calendar of North China.

caused by the bidirectional reflectance distribution function (BRDF) composites, still remained. It was necessary to reduce the remaining noise for further analysis. In this study, we downloaded MOD09A1 data for 2004 (forty-six 8-day composites) from the USGS EROS Data Centre (<http://www.edc.usgs.gov>).

3.2. Vegetation index

NDVI is an index that shows the absorptive and reflective characteristics of vegetation in the red and near infrared portions of the electromagnetic spectrum. Therefore, changes in the NDVI time-series indicate changes in vegetation conditions proportional to the absorption of photosynthetically active radiation (Sellers, 1985). Therefore, we used NDVI in our method. For each 8-day composite, we calculated NDVI, using surface reflectance values from the red, NIR (814–875 nm) bands (Eq. (1)):

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{red}}}{\rho_{\text{NIR}} + \rho_{\text{red}}} \quad (1)$$

3.3. Validation data set

In order to assess the crop map, we gathered national agricultural statistical data to compare with the satellite-derived crop area estimates. The statistical crop data area at county and province-level were obtained from the Chinese Ministry of Agriculture database.

Trimble GeoXH hand-held GPS receivers were used for the ground survey in 2005, and a set of 69 field boundaries were sampled (Fig. 1). There was only one type of crop in each field. Every field was about 1 km². The field polygons were spatially referenced to the satellite data set. These polygons were used to label unsupervised clusters in the land cover classification.

3.4. Pre-processing

Even though the available NDVI data sets were corrected for gaseous and aerosol scattering, the thick clouds still remained as noise (Los et al., 1994). To remove data affected by thick clouds, we extracted the information on clouds and generated masks of cloud cover for all time periods of each MODIS tile using the quality control flags in the MODIS file. Those pixels, labelled as clouds, were removed. For those pixels that were not labelled as clouds in the cloud quality flag, an additional restriction was then applied. The pixels with a blue reflectance of ≥ 0.2 were removed as abnormal data (Xiao et al., 2005).

Geometric, radiometric and atmospheric corrections had been made to the MODIS data used here. Data was re-sampled systematically to a uniform spatial resolution (500 m), and then were re-projected from the sinusoidal (SIN) projection to the standard UTM projection.

To use the FFT, observations must be relatively error (cloud) free and equidistant in time. The algorithms of masking clouds always result in an image with data gaps. It is necessary to reconstruct a cloud free profile of NDVI for further performance. In this work, the Savitzky–Golay filter is used to smooth out noise in NDVI time-series. Compared to other methods, the method based on Savitzky–Golay filter takes advantage of ancillary data in the form of cloud flags and can reconstruct high-quality NDVI time-series by setting only two parameters (Chen et al., 2004). In addition, it is a very simple theory and easy to implement. The parameters in the Savitzky–Golay filter must be determined according to the NDVI observations when the filter is applied to NDVI time-series smoothing. The first parameter is m , the half-width of the smoothing window. The second parameter is an integer (d) specifying the degree of the smoothing polynomial. The (m , d) combination of (7, 3) was determined for Savitzky–Golay filter in long-term change trend fitting process. The (m , d) combination of (4, 6) was determined in the fitting iteration process.

3.5. Fourier transformation

Crops exhibit distinctive seasonal patterns. The NDVI series data can be used to dynamically reflect crop growth and track crop phenological metrics change, because of the positive correlation between NDVI and LAI. Amplitude and phase angle images, derived from Fourier analysis of the time-series NDVI data, were correlated with information on crop type (Jakubauskas et al., 2001).

The function $f(t)$ is transformed in the discrete Fourier transform as follows:

$$F(k) = \frac{1}{N} \sum_{t=0}^{N-1} f(t) \exp\left(-\frac{2\pi i k t}{N}\right), \quad (2)$$

where $k = 0, 1, 2, \dots, N - 1$, and N is the total number of input data.

In this work, we used Fourier transform subroutines implemented by the Interactive Data Language (IDL: Research Systems Inc., Boulder, USA).

The FFT was applied to the NDVI data sets on a per pixel basis for the entire study area. The real and

imaginary components of the Fourier transformation are described as amplitude and phase, represented in units of NDVI and time, respectively. Table 1 shows the first four terms of Fourier analysis for four main cropping systems in northern China. The phase value ranged from 0 to 2π . The ‘g’ and ‘Cum.g’ are the contributions to the total variance (Table 1). Most crop information is in terms 0, 1, 2, and 3 (Table 1). The amplitudes of terms 0, 1, 2, 3 and the phases of terms 1, 2, 3 are selected as classification parameters for distinguishing the fields of winter wheat–maize and winter wheat–cotton from areas with double cropping systems. The other terms were discarded because they had trivial amplitudes.

3.6. Classification

The National Land Cover Project (NLCD), supported by the Chinese Academy of Sciences, completed the analysis of the Landsat 7 Enhanced Thematic Mapper (ETM+) images acquired in 1999 and 2000 for all of China (Liu et al., 2003). 508 ETM+ images in 1999/2000 were geo-referenced and ortho-rectified, using field collected ground control point and time-resolution digital elevation models. A classification system of 25 land cover types was used in the NLCD project. Visual interpretation of ETM+ images was

conducted to generate a thematic map of land cover in China at a scale of 1:100,000. The resulting vector data set was converted into a gridded database at 1 km spatial resolution.

Based on the national land use/land cover vector data (1:100,000), we produced a 500 m resolution grid cell set with the majority rule (Benson and MacKenzie, 1995). The cropland masks were generated with the land use/land cover map, and the cropland pixels of NDVI data were extracted.

In this study, multi-stage classification was used to discriminate crop types. Firstly, unsupervised classification, based on amplitudes of terms 0, 1, 2, 3, was produced using the ISODATA clustering algorithm and ERDAS 8.5 software. Thirty clusters were produced and merged to a set of two cropland formations: fields with one cropping system and areas with double cropping systems. Cluster merging and labelling were based on the set of reference polygons. Secondly, supervised classification was performed using the maximum likelihood decision rule and ERDAS 8.5 software. The amplitudes of terms 0, 1, 2 and phases of term 1, 2 were selected as classified parameters for distinguishing fields of spring maize and cotton from areas with one cropping system. The amplitudes of terms 0, 1, 2, 3 and the phases of terms 1, 2, 3 were selected as classification parameters for distinguishing

Table 1
Parameters of Harmonic function for crops

	Term	Amplitude	Phase	g	Cum.g
Rotation of winter wheat–maize	0	48.67			
	1	8.99	2.77	0.25	0.25
	2	8.62	−2.17	0.24	0.49
	3	8.06	1.19	0.22	0.71
	4	0.96	−1.89	0.03	0.73
Relay intercropping of winter wheat–cotton	0	48.52			
	1	5.20	2.96	0.19	0.19
	2	3.52	−2.17	0.13	0.32
	3	10.40	1.43	0.38	0.70
	4	2.73	3.11	0.10	0.80
Single cropping system of cotton	0	40.35			
	1	18.52	2.45	0.53	0.53
	2	8.95	−1.30	0.25	0.78
	3	2.46	1.92	0.07	0.85
	4	1.01	−0.73	0.03	0.88
Single cropping system of spring maize	0	32.98			
	1	15.14	2.52	0.52	0.52
	2	8.70	−1.26	0.30	0.82
	3	2.84	1.33	0.10	0.92
	4	0.10	0.83	0.00	0.92

$$g_i = \frac{\text{Amplitude}_i}{\sum \text{Amplitude}_j} \quad (i = 1, 2, 3, 4 \quad j = 1, 2, 3, 4 \dots), \quad \text{Cum.g}_i = \sum_{j=1}^i g_j \quad (i = 1, 2, 3, 4)$$

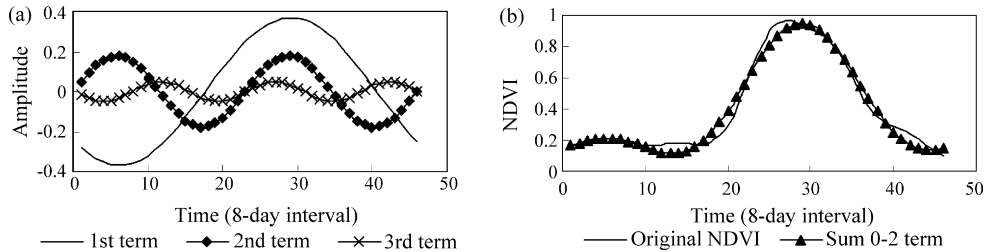


Fig. 3. First three-term curves for cotton fields (a). Original NDVI profile for cotton and NDVI profile constructed from the summing harmonic 0, 1, 2, 3 (b).

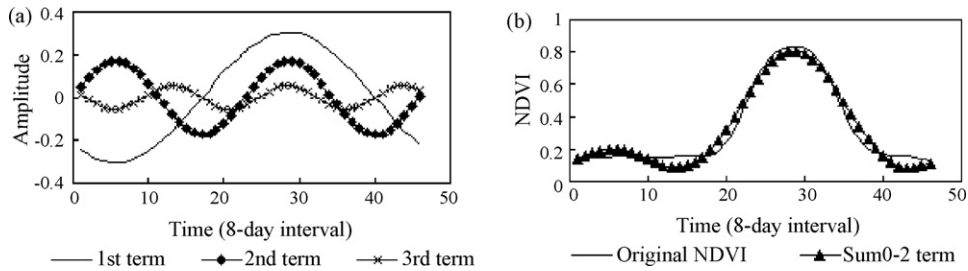


Fig. 4. First three-term curves for fields of spring maize (a). Original NDVI profile for spring maize and NDVI profile constructed from the summing harmonic 0, 1, 2, 3 (b).

the fields of winter wheat–maize and winter wheat–cotton from areas with double cropping systems. The spectral signatures were defined based on the set of 69 reference data.

4. Results

4.1. Feature analysis of NDVI series curve

Before discriminating crop types, an extensive field survey was performed throughout the area of northern China using Global Positioning System (GPS) equipment. A total of 69 polygons were obtained from the field survey from which 4 ground truth polygons were used as reference data to extract representative pixels. The representative pixels dominated by one cropping

system. The averaged value of the pixels is used as reference NDVI patterns.

The NDVI profile for fields with one cropping system has one modal shape (Figs. 3(b) and 4(b)). Areas with double cropping systems have a trimodal shape (Figs. 5(b) and 6(b)). The number of modals is an important characteristic to use to separate fields with one cropping system and double cropping systems. Amplitude of term 1 in areas with one cropping system is greatly larger than other terms' excluding term 0 (Figs. 3(a) and 4(a)). In areas with double cropping systems, amplitude of terms 1, 2, 3 is relatively larger than term 0 (Figs. 5(a) and 6(a)). Therefore, unsupervised classification, based on amplitudes of terms 0, 1, 2, 3, was particularly suited to discriminate fields with a one cropping system from fields with double cropping

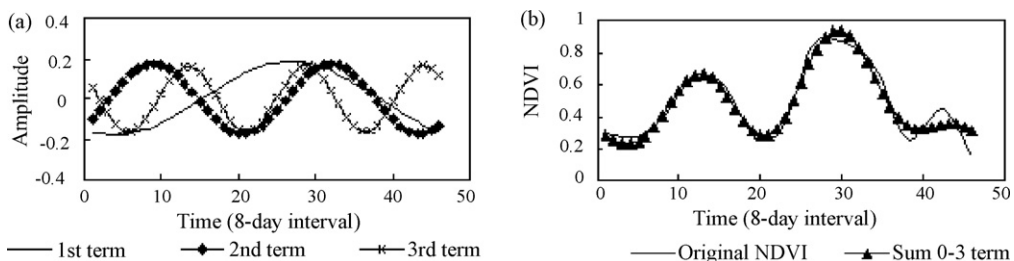


Fig. 5. First three-term curves for fields of winter wheat–cotton (a). Original NDVI profile for winter wheat–cotton and NDVI profile constructed from the summing harmonic 0, 1, 2, 3 (b).

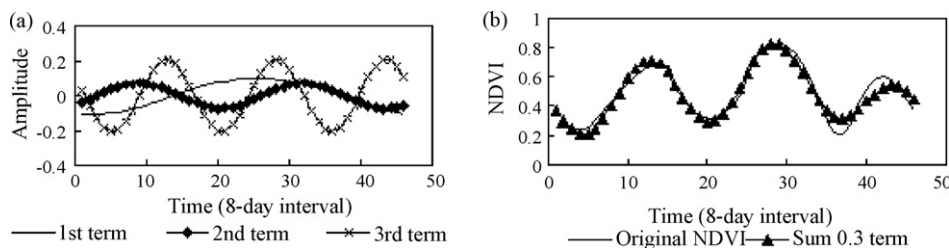


Fig. 6. First three-term curves for fields of winter wheat–maize (a). Original NDVI profile for winter wheat–maize and NDVI profile reconstructed from the summing harmonic 0, 1, 2, 3 (b).

systems. Performed after stepwise unsupervised classification, supervised classification was only performed on a small subspace of the total spectral heterogeneity. Thus, the strength of supervised classification could be realistically utilized to classify crop types.

Spring maize and cotton are the main crops in areas which use one cropping system. The amplitude of 0th

term for cotton in areas with a one cropping system is greater than spring maize (Table 1), because spring cotton has a longer growth period. The peak of spring cotton's NDVI occurs earlier than spring maize. Moreover, the value of spring cotton at peak time is greater than spring maize (Figs. 3(b) and 4(b)). This is present during the 1st and 2nd terms (Figs. 3(a) and

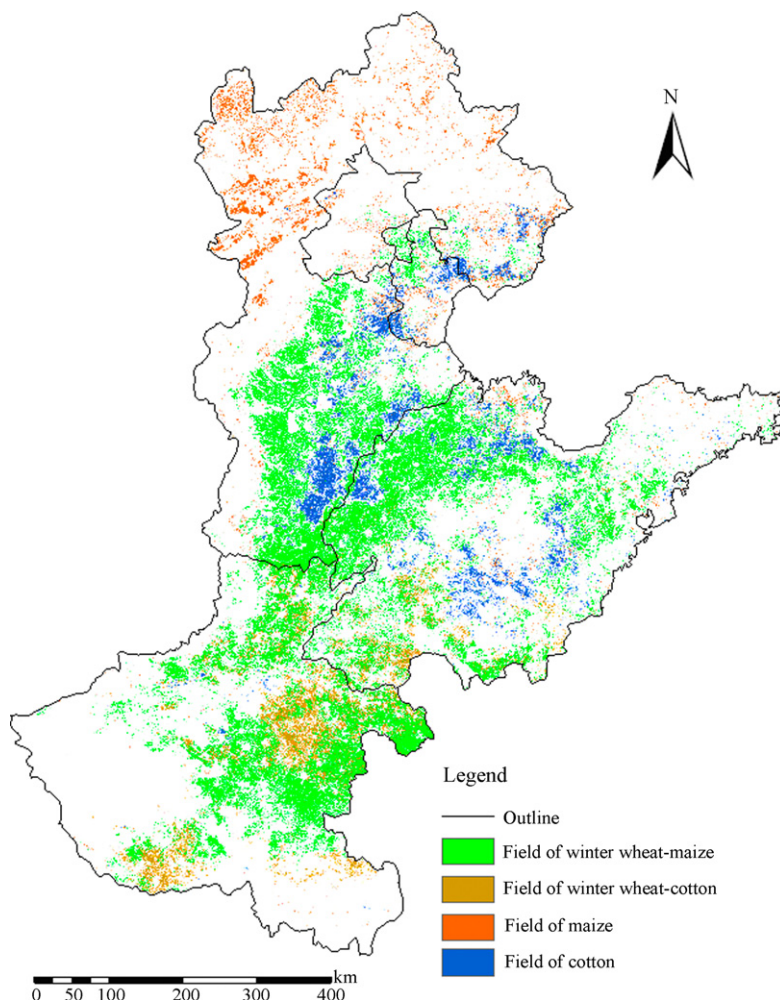


Fig. 7. Distribution of maize and cotton in North China in 2004.

Table 2

A provincial-level comparison of area estimates of crop fields (km²) from MODIS data and statistical data

Province or municipality	Statistics (km ²)		Estimated using MODIS data (km ²)		The agreement (%) using statistical data set as basis	
	Maize	Cotton	Maize	Cotton	Maize	Cotton
Hebei	26305.70	6691.33	29004.80	7234.89	90	92
Beijing	935.40	65.40	637.29	44.65	68	69
Tianjin	1347.55	868.67	1296.04	1151.80	96	67
Shandong	24550.50	10592.07	28187.00	10231.79	85	97
Henan	24200.00	9518.00	28476.13	11303.31	82	81

4(a)). These characteristics are useful for distinguishing between spring maize and spring cotton. Therefore, amplitudes of terms 0, 1, 2 and phases of term 1, 2 can be used to identify fields of spring maize and cotton in areas with a one cropping system.

In areas with a double cropping system, winter wheat–summer maize and winter wheat–summer cotton are the most important crop systems. Summer cotton needs to be pre-bred before the winter wheat is harvested in May. When summer cotton is planted, summer maize is seeded. Therefore, cotton grows quickly early on in the growth cycle. This is described by the phase of 2nd term (Figs. 5(a) and 6(a)). Summer cotton needs more time to finish its period and seeding winter wheat is delayed (Figs. 5(b) and 6(b)). Therefore, amplitude of terms 0, 1, 2, 3 and phase of terms 1, 2, 3 for winter wheat–cotton is different from winter wheat–summer maize.

4.2. Classification results

Results (Fig. 7) show that fields of winter wheat–maize are mainly distributed in the southeast of HEBEI province, the northwest of SHANDONG province and the northeast of HENAN province, while fields of maize are mainly distributed in the northern area of HEBEI province.

The fields of winter wheat–cotton are mainly distributed in the southwest of the SHANDONG province and the south and east of the HENAN province, while cotton is mainly distributed in the center and eastern parts of the HEBEI province, and some parts of the TIANJIN and SHANDONG provinces.

4.3. Comparison with statistics on crop acreage

We assembled agricultural statistical data to compare to the MODIS-derived crop acreage for crop area prediction. The MODIS-based crop map was evaluated

in two ways: (1) province-level comparison (Table 2), and (2) county-level comparison (Fig. 8(a and b)). Agricultural census data usually reports the total area of cotton or maize fields. Therefore, the purpose of the comparison between MODIS-derived crop map and statistical data was to evaluate the total cropland area, sown with maize or cotton, which was estimated by MODIS data.

Table 2 shows the difference between the estimated acreage of the two crop areas and the statistics at province-level. The maize and cotton area in Beijing were greatly underestimated. The cotton area in Tianjin was greatly overestimated. There was little difference

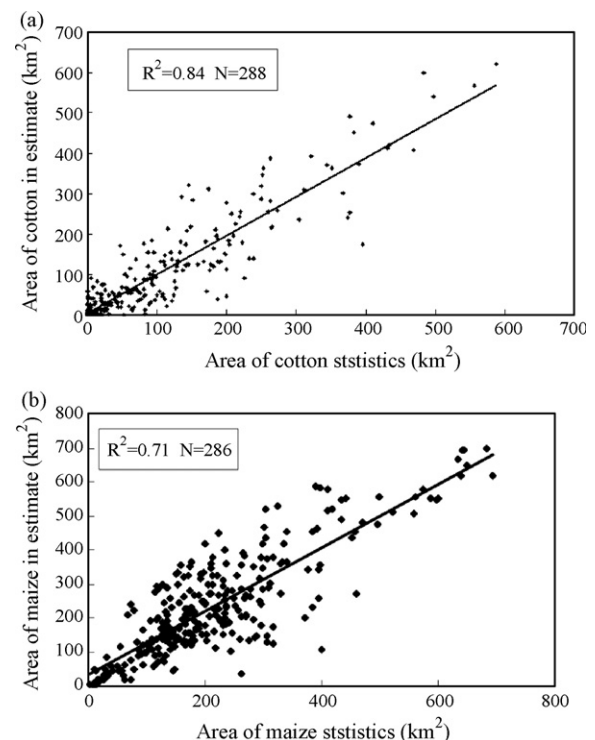


Fig. 8. A county-level comparison of crops (cotton (a), maize (b)) area (km²) from MODIS data and statistical data.

between the MODIS-derived area and the statistics of the two crops in the three provinces.

At county-level, the estimated cotton area and statistical data were well correlated ($R^2 = 0.84$, RMSE = 48.11 km²) (Fig. 8a). The correlation in maize areas from the MODIS data and the statistical data was positive ($R^2 = 0.71$, RMSE = 82.00 km²) (Fig. 8b).

5. Discussion and conclusions

In this study, we applied the FFT analysis of MODIS-derived vegetation indices to an area with a double cropping system and discriminated maize and cotton fields at regional spatial scales. The estimated crop (cotton and maize) areas by MODIS data were well correlated with statistical data at county-level. The crop map overestimated or underestimated areas of cotton and maize due to the fragmentation and sub-pixel proportion of cotton or maize fields at county-level. If finer spatial resolution data could be used, further improvements to crop identification could be achieved. At the province-level, MODIS-derived crop (cotton, maize) areas corresponded with statistical data, excluding Beijing and Tianjin. The maize and cotton areas in Beijing were greatly underestimated. The cotton area in Tianjin was greatly overestimated. The small fields and various types of planting structure maybe the reason that it is difficult to estimate Beijing and Tianjin's crop areas. Our study shows that the method, using MODIS NDVI time-series, is promising for crop mapping in northern China with double cropping systems. Finally, a FFT analysis of the NDVI profile needs a full cycle of data.

Acknowledgments

This research was supported by the program “Regional crop growth monitoring and simulation based on multi-sensor information and assimilation of remote sensing data into crop growth model” from the National High Technology Research and Development Program of China (863 Program No. 2006AA12Z103) and by the Program “Research on data fusion and its criterion for agricultural resources spatial information” from National Key Technologies R&D Program of China (No. 2006BAD10A06) and by the Asia ITC project “HUABEI-CGMS: Crop Growth Monitoring and Yield Forecasting in the North China plain (Huabei)” from the European Union. We are also grateful to Reviewers and editor for their comments and suggestions, which have greatly help us to improve the original version of the manuscript.

References

- Azzali, S., Menenti, M., 2000. Mapping vegetation–soil–climate complexes in southern Africa using temporal Fourier analysis of NOAA-AVHRR NDVI data. *Int. J. Remote Sens.* 21, 973–996.
- Brown, J.F., Loveland, T.R., Merchant, J.W., Reed, B.C., Ohlen, D.O., 1993. Using multi-source data in global land cover characterization: concepts, requirements and methods. *Photogramm. Eng. Remote Sens.* 59, 977–987.
- Chen, J., Jonsson, P., Tamura, M., Gu, Z., Matsushita, B., Eklundh, L., 2004. A simple method for reconstructing a high-quality NDVI time-series data set based on the Savitzky–Golay filter. *Remote Sens. Environ.* 91, 332–344.
- Defries, R., Hansen, M., Townshend, J., 1995. Global discrimination of land cover types from metrics derived from AVHRR Pathfinder data. *Remote Sens. Environ.* 54, 209–222.
- Doraiswamy, P.C., Sinclair, T.R., Hollinger, S., Akhmedov, B., Stern, A., Prueger, J., 2005. Application of MODIS derived parameters for regional crop yield assessment. *Remote Sens. Environ.* 97, 192–202.
- Eastman, J.R., Fulk, M.A., 1993. Long sequence time series evaluation using standardized principal component. *Photogramm. Eng. Remote Sens.* 59, 1307–1312.
- Gutman, G., Ignatov, A., 1998. Derivation of green vegetation fraction from NOAA-AVHRR for use in numerical weather prediction models. *Int. J. Remote Sens.* 19, 1533–1543.
- Jakubauskas, M.E., Legates, D.R., Kastens, J.H., 2001. Harmonic analysis of time-series AVHRR NDVI data. *Photogramm. Eng. Remote Sens.* 67, 461–470.
- Jakubauskas, M.E., Legates, D.R., Kastens, J.H., 2002. Crop identification using harmonic analysis of time-series AVHRR NDVI data. *Comput. Electron. Agric.* 37, 127–139.
- Liu, J., Liu, M., Zhang, D., Deng, X., 2003. Study on spatial patterns of land use change in China during 1995–2000. *Sci. China* 46, 373–384.
- Los, S.O., Justice, C.O., Tucker, C.J., 1994. A global 1 by 1 degree NDVI data set for climate studies derived from the GIMMS continental NDVI data. *Int. J. Remote Sens.* 15, 3493–3518.
- Loveland, T.R., Merchant, J.W., Brown, J.F., Ohlen, D.O., Reed, B.C., Olson, P., Hutchinson, J., 1995. Seasonal land-cover regions of the United States. *Ann. Assoc. Am. Geographers* 85, 339–355.
- Menenti, M., Azzali, S., Verhoef, W., Van Swol, R., 1993. Mapping agroecological zones and time lag in vegetation growth by means of Fourier analysis of time series of NDVI images. *Adv. Space Res.* 13, 233–237.
- Reed, B.C., Brown, J.F., VanderZee, D., Loveland, T.R., James, M.W., Ohlen, D.O., 1994. Measuring phenological variability from satellite imagery. *J. Veg. Sci.* 5, 703–714.
- Roerink, G.J., Menenti, M., 2000. Reconstructing cloudfree NDVI composites using Fourier analysis of time series. *Int. J. Remote Sens.* 21, 1911–1917.
- Sellers, P.J., 1985. Canopy reflectance photosynthesis and transpiration. *Int. J. Remote Sens.* 6, 1335–1372.
- Townshend, J.R., Justice, C.O., Kalb, V.T., 1987. Characterization and classification of South American land cover types using satellite data. *Int. J. Remote Sens.* 8, 1189–1207.
- Townshend, J., Justice, C., Li, W., Gurney, C., McManus, J., 1991. Global land cover classification by remote sensing: present capabilities and future possibilities. *Remote Sens. Environ.* 35, 243–255.
- Tucher, C.T., Townshend, J.R., Goff, T.E., 1985. African land cover classification using satellite data. *Science* 227, 369–375.

- Vermote, E.F., Vermeulen, A. (1999). MODIS Algorithm Technical Background Document, Atmospheric correction algorithm: Spectral reflectances (MOD09). NASA contract NAS5-96062.
- Xiao, X., Boles, S., Liu, J., Zhuang, D., Froking, S., LI, C., Salas, W., Moore III, B., 2005. Mapping paddy rice agriculture in southern China using multi-temporal MODIS images. *Remote Sens. Environ.* 95, 480–492.
- Yin, Z.W.T.H.L., 1997. Obtaining spatial and temporal vegetation data from Landsat MSS and AVHRR/NOAA satellite images for a hydrologic model. *Photogramm. Eng. Remote Sens.* 63, 69–77.
- Zhang, X., Friedl, M.A., Schaaf, C.B., Strahler, A.H., Hodges, J.C.F., Gao, F., Reed, B.C., Huete, A., 2003. Monitoring vegetation phenology using MODIS. *Remote Sens. Environ.* 84, 471–475.